

# Deep Learning Model Development and Evaluation

## Project Title: Automated Pneumonia Detection from Chest X-Ray Images using EfficientNet-B3

Student Name: A. K. M. Thohidul Alam Khan

Student ID: 01

Course Name: AIDL

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## 1. Problem Definition

### 1.1 Problem Statement

Pneumonia is a severe respiratory infection that inflames the air sacs in one or both lungs, often causing fluid buildup. Early and accurate diagnosis is critical for effective patient treatment. Traditionally, radiologists diagnose pneumonia by visually inspecting Chest X-Ray (CXR) scans. However, this manual process is time-consuming, prone to human error due to fatigue, and requires highly trained medical professionals who are often scarce in rural or developing areas.

The objective of this project is to develop an automated binary classification system that can analyze Chest X-Ray images and accurately classify them as either **NORMAL** (healthy) or **PNEUMONIA** (infected).

### 1.2 Justification for Deep Learning

Deep learning, specifically Convolutional Neural Networks (CNNs), is exceptionally well-suited for this task because:

- **Spatial Feature Extraction:** CNNs can automatically learn hierarchical spatial features (like edges, textures, and fluid opacities) directly from raw pixel data without manual feature engineering.
- **High Accuracy in Medical Imaging:** Transfer learning models pre-trained on large datasets (like ImageNet) have repeatedly achieved state-of-the-art results in medical image analysis.

### 1.3 Expected Outcomes

- A robust model with high **Recall (Sensitivity)** to ensure that patients who actually have pneumonia are not misclassified as healthy (minimizing False Negatives).
- A functional inference pipeline that can process unseen X-Ray images.
- Explainability features (Heatmaps) to show exactly which region of the lungs led to the AI's diagnosis.

## 2. Dataset

### 2.1 Dataset Characteristics

The project utilizes the publicly available **Kaggle Chest X-Ray Images (Pneumonia)** dataset.

- **Total Images:** 5,840 JPEG images.
- **Classes:** 2 classes — **NORMAL** and **PNEUMONIA**.
- **Distribution Challenge:** The dataset is inherently imbalanced. During the data splitting phase, the training subset contained approximately **1195 NORMAL** and **3500 PNEUMONIA** cases.

### 2.2 Data Preprocessing and Augmentation

The `src/dataset.py` module handles a robust data pipeline:

1. **Grayscale to RGB:** All X-rays (mode 'L') were converted to 3-channel RGB to match the input requirements of pretrained ImageNet models.
2. **Resizing:** Images were resized to **224x224** pixels.
3. **Normalization:** Standard ImageNet mean **[0.485, 0.456, 0.406]** and standard deviation **[0.229, 0.224, 0.225]** were applied.
4. **Data Augmentation:** To prevent overfitting and generalize better, the training set was augmented using:
  - Random Horizontal Flips ( $p=0.5$ )
  - Random Rotation (**10** degrees)
  - Color Jitter (Brightness and Contrast tweaks)
  - Minor Affine Translations (**5%**)
5. **Class Imbalance Handling:** A `WeightedRandomSampler` was implemented to assign higher sampling weights to the minority **NORMAL** class, ensuring the model does not become biased towards predicting **PNEUMONIA**.

## 3. Model Development

### 3.1 Chosen Architecture

The chosen architecture for this project is **EfficientNet-B3**.

### 3.2 Justification for the Model

EfficientNet was chosen over older models like VGG-16 or ResNet-50 because it uses a **Compound Scaling** method. It uniformly scales network width, depth, and resolution in a principled way.

- Parameter Efficiency:** EfficientNet-B3 achieves state-of-the-art accuracy with significantly fewer parameters (**12 million**) compared to VGG-16 (138 million) or ResNet-50 (~25 million).
- Medical Imaging Fit:** It is highly capable of capturing both fine-grained details (small lung opacities) and macro-structures (rib cages) efficiently.

### 3.3 Model Modifications

The model was imported from `torchvision.models` with pre-trained ImageNet weights. The base architecture was frozen, and the default 1000-class classifier was replaced with a custom binary classification head:

- `Dropout(p=0.3)` layer to prevent overfitting.
- `Linear(in_features=1536, out_features=1)` layer. A single output logit is used in conjunction with `BCEWithLogitsLoss` for numerical stability.

## 4. Implementation

### 4.1 Framework and Tools

- Deep Learning Framework:** PyTorch
- Experiment Tracking:** MLflow
- API & Deployment:** FastAPI, Uvicorn
- Explainability:** PyTorch Grad-CAM

### 4.2 Reproducibility

To ensure the training process is reproducible, a global random seed `42` was enforced across PyTorch, NumPy, and DataLoaders (`torch.manual_seed(42)`).

## 5. Training Process

### 5.1 Dataset Splitting

The original dataset provided a tiny validation set (16 images) which is statistically insignificant. To solve this, the original `train` directory was dynamically split into:

- Training Set:** 90% (4,695 images)
- Validation Set:** 10% (521 images)
- Test Set:** Original test directory (624 images)

### 5.2 Hyperparameters & Loss Function

- Batch Size:** 32
- Epochs:** Maximum 50 (with Early Stopping patience of 7 epochs)
- Optimizer:** Adam Optimizer
- Learning Rate:** 0.001 for the classifier head.
- LR Scheduler:** CosineAnnealingLR
- Loss Function:** `nn.BCEWithLogitsLoss` equipped with a dynamic `pos_weight` parameter to further penalize the model for missing minority class instances.

## 6. Performance Evaluation

The model was evaluated on the unseen Test Set (624 images). The primary goal was to maximize **Recall** for the pneumonia class.

### 6.1 Quantitative Metrics

The model achieved the following results on the test set:

- Accuracy:** `0.8814` (88.14%)
- Precision:** `0.9365` (93.65%)
- Recall (Sensitivity):** `0.8692` (86.92%)

- **F1-Score:** 0.9016 (90.16%)
- **AUC-ROC:** 0.9455 (94.55%)

Detailed Classification Report:

|           | precision | recall | f1-score | support |
|-----------|-----------|--------|----------|---------|
| NORMAL    | 0.81      | 0.90   | 0.85     | 234     |
| PNEUMONIA | 0.94      | 0.87   | 0.90     | 390     |

## 6.2 Visualizations

### ROC Curve & PR Curve

The AUC-ROC score of **0.9455** indicates an excellent capability of the model to distinguish between healthy and pneumonia-infected lungs across various threshold levels.

## 7. Results and Analysis

### 7.1 Interpretation of Results

The model performed exceptionally well, achieving an F1-score of over 90%.

- **Strengths:** The precision is extremely high (93.65%). This means when the model diagnoses a patient with Pneumonia, it is highly trustworthy. The AUC-ROC of 0.9455 proves the model's predictive power is far beyond random guessing.
- **Limitations:** The recall is slightly lower at 86.92%. In a medical context, missing a sick patient (False Negative) is riskier than a False Positive. While 87% is strong for a Stage-1 frozen-backbone training phase, there is room for improvement to push this closer to 95%.

### 7.2 Baseline Comparison

The project source code includes a custom-built 3-block CNN ( `baseline_cnn` ). Comparing the baseline CNN with the transfer-learning EfficientNet-B3 model demonstrates the power of transfer learning. While a scratch-built CNN typically struggles to break 80-85% accuracy on this imbalanced dataset without extensive tuning, the EfficientNet model immediately achieved ~88% accuracy and 94% AUC-ROC.

## 8. Improvement Strategy & Innovations

### 8.1 Fine-Tuning (Stage 2)

Currently, the model was trained by freezing the backbone and only training the classifier head. To improve the Recall metric, the next step is **Fine-Tuning**. The codebase has already been equipped with a `--finetune` flag which unfreezes the last three blocks of the EfficientNet-B3 model and uses a differential learning rate (e.g., `1e-5` for the backbone, `1e-4` for the head) to adapt the ImageNet weights specifically to X-ray textures.

### 8.2 Innovation 1: Explainable AI (Grad-CAM)

To build trust with medical professionals, **Gradient-weighted Class Activation Mapping (Grad-CAM)** was integrated. Instead of acting as a "black box," the model generates a heatmap overlay on the input X-Ray, visually highlighting the fluid opacities or lung regions that triggered the pneumonia classification.

### 8.3 Innovation 2: Local Web Deployment

A complete end-to-end production pipeline was established:

- **FastAPI Backend:** Provides robust API endpoints for inference.
- **Glassmorphism Frontend UI:** A locally deployed web application that allows users to upload an X-ray via a graphical interface, view the AI's confidence score dynamically, and instantly see the Grad-CAM heatmap diagnosis.